Ramesh K P Senior Embedded Engineer

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Perinthalmanna, Malappuram, Kerala, 679322

Summary

Embedded Firmware Developer with 10+ years of experience designing and delivering reliable embedded systems across automotive, industrial automation, and IoT domains. Skilled in Embedded C, CAN, I2C, SPI, UART, and MQTT communication protocols with hands-on expertise on PIC, ARM, and ESP32 microcontrollers. Proven track record in firmware architecture, diagnostic testing (UDS, SIL), and system optimization. Adept at driving cross-functional collaboration and delivering high-quality embedded solutions within timelines.

SKILLS

Languages
Embedded C, Python, Assembly Language

Microcontrollers
PIC (8/16/32-bit), ARM, AVR, ESP32, Arduino, Raspberry Pi

Tools/IDEs
MPLAB X, VS Code, CCS, Arduino IDE, CANoe, Git, Gerrit, Jira, SonarCloud, CMake, GCC, GDB, JTAG, Kvaser

Hardware Design
Circuit design, PCB layout (EAGLE, Proteus), schematic validation

Other Skills
IoT, Basics of Embedded Linux, RTOS, basic OpenCV, C++, and MATLAB

Firmware Development
BareMetal, Firmware architecture, Device driver development, Low level hardware control.

Communication Protocols
UART, I2C, SPI, Modbus, CAN (J1939, UDS), RS232/RS485, MQTT

Testing & Debugging
SIL, CAPL scripting, ECU flashing, GTest, Python mocks, Static code analysis

Methodologies
Agile, Behavior-Driven Development (BDD) with Gherkin, CI/CD pipelines

Work Experience

06/2023 – 09/2025
Kochi

Senior Embedded Software Engineer
KPIT Technologies Limited, Kochi

- **Project 1: Fuel Cell Electrical Bus System & Marine Vehicle ECU Development**
  - Developed Embedded C firmware for ECUs managing fuel-cell propulsion systems in marine applications.
  - Implemented multi-protocol communication (CAN [J1939], I2C, RS485) for real-time data exchange and subsystem coordination.

- Designed a robust CAN communication stack with error detection, fault recovery, and retransmission mechanisms.
    - Performed low-level firmware–hardware optimization through schematic analysis and on-board debugging.
    - Executed secure ECU flashing with version control and rollback support to ensure safe updates.
    - Validated communication reliability and timing through hardware analyzers and diagnostic testing.
- **Project 2: SIL, Diagnostic Testing & Automation for Automotive ECUs**
    - Conducted Software-in-the-Loop (SIL) validation using virtual ECUs to test logic and diagnostic functionality.
    - Utilized CANoe, Corvus, and UDS protocols integrated with BDD (Gherkin) for automated diagnostic testing.
    - Developed Python-based automation to execute CAN diagnostic sequences and regression testing.
    - Authored and verified CAPL scripts simulating real-time CAN communication scenarios.
    - Analyzed CAN signals for protocol compliance, timing accuracy, and message integrity.
- **Project 3: Network Communication & Display Software for Automotive Clusters**
    - Implemented bidirectional TCP communication between host systems and Docker-based AGL clusters.
    - Developed AGL display modules for vehicle-speed and gear visualization in real-time.
    - Configured CMake and GCC cross-compilers for efficient cross-platform builds and deployment.
    - Conducted unit and mock testing using GTest and Python automation frameworks.
    - Debugged and profiled runtime performance to ensure optimized display responsiveness.

08/2021 – 05/2023
Kochi

Senior Software Engineer
Naico ITS, Kochi

- **Project 1: Smart Water Quality Monitoring and Control System**
    - Lead the project team through end-to-end system design, code review and analysis, and regular stakeholder alignment meetings to ensure technical and functional objectives were met.
    - Lead firmware architecture and development for IoT-enabled industrial automation controllers.
    - Integrated multi-sensor inputs (pH, TDS, EC, Turbidity, COD, BOD) using MODBUS and UART for cloud monitoring.
    - Implemented MQTT-based data publishing and low-power firmware design for remote field devices.
    - Automated data analytics and report generation through Python scripts for real-time insights.
    - Enforced code quality and CI/CD pipelines using SonarCloud, GitLab, and JIRA for traceable releases.
    - Delivered a scalable, secure, and compliant automation solution meeting industrial standards.

- **Project 2: Multi-Modal Biometric Attendance & Access System**
  - Designed embedded firmware for an integrated biometric platform combining face, fingerprint, and RFID.
  - Deployed Raspberry Pi as an edge controller for gate automation and real-time access decisions.
  - Integrated fingerprint sensors and RFID modules with embedded security layers.
  - Managed encrypted data storage, timestamped logging, and cloud synchronization for accuracy.

02/2020 – 05/2020
Bangalore

Embedded Engineer
Divistha Networks, Bangalore

- **Project: Linux Display Architecture and Software Testing**
  - Gained practical expertise in Linux display subsystems (Frame Buffer, X11, X Server).
  - Performed regression and interface testing ensuring visual consistency across platforms.
  - Designed test scenarios for GUI performance benchmarking and validation.
  - Collaborated with developers to resolve software and driver integration issues.

08/2018 – 11/2019
Trivandrum

Embedded Engineer
Costech, Trivandrum

- **Project: Web-Enabled Remote Display System (WeRD)**
  - Directed the project team in architecture design, static code analysis, and periodic stakeholder review sessions to maintain development quality and project alignment.
  - Developed firmware for web-enabled LED display modules (P10, P3) with remote configuration capability.
  - Created custom C device drivers integrating UART, SPI, I2C, and RS232 interfaces.
  - Lead full SDLC activities—from architecture and coding to deployment—for embedded displays.
  - Optimized refresh timing and data throughput to improve visual performance.
  - Authored test cases and validation documentation ensuring system reliability.

09/2014 – 07/2016
Kochi

Embedded Engineer
Focuz Infotech, Kochi

- Developed IoT-enabled systems using PIC, ARM, and AVR microcontrollers for real-time monitoring.
- Integrated PWM and ADC modules and optimized power management.
- Utilized MQTT for cloud connectivity and remote management.
- Mentored junior engineers and enforced C coding standards.
- Executed system testing and debugging ensuring reliability.

12/2012 – 05/2014

Senior Embedded Engineer
Cutesys Technologies

- Designed embedded solutions using 8-/16-/32-bit controllers and implemented data-logging modules.
- Executed PCB layout and hardware design in collaboration with electronics teams.
- Optimized UART, I2C, and ADC communication for low-power operation.
- Integrated safety-compliance mechanisms meeting industrial standards.
- Improved system responsiveness and long-term stability through code optimization.

10/2011 – 11/2012
Kochi

Junior Embedded Engineer
Grey Technolabs, Kochi

- Developed firmware on 16-bit PIC controllers emphasizing performance and scalability.
- Designed structured Embedded C code aligned with best practices.
- Trained students and faculty in embedded systems programming.
- Guided entry-level engineers in debugging and testing workflows.
- Performed static code analysis for quality and compliance.
- Conducted comprehensive validation to ensure robustness and reliability.

Education

Tamilnadu

Bsc: Mathematics
Bharathiyar University

Perintalmanna

Higher Secondery
Sree Valluvanad Vidhya Bhavan

Languages

English — Fluent

Hindi — Basic

Tamil — Conversational

Malayalam — Native/Bilingual

Interests

- Chess
- Traveling
- Music